

Result: 67% 8 / 12	 101710 000226	Neptun ID: Name: _____ Draft papers: ☐
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The test consists of **4 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

1.) (essay)

2.) (essay)

3.)

a	b	c	d
☐	☐	☐	☐

4.)

☐	☐	☐	☐
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Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) When can we say that a set of formulae is satisfiable?

Score: 2 / 2.

2.) Define the conjunctive normal form (CNF).

Score: 2 / 2.

3.) Which statements are true for the following three formula?

$$A \Leftrightarrow \neg q, \quad B \Leftrightarrow \neg q, \quad C \Leftrightarrow ((p \equiv q) \supset q) \wedge p \supset q$$

2 $A \Leftrightarrow B,$

0 $\{A, B, C\}$ is satisfiable,

(c) $A, B \models C,$

(d) $(A \vee B)$ is valid.

4.) Which statements are true for the following three formula?

$$A \Leftrightarrow \neg q, \quad B \Leftrightarrow (q \supset p) \supset q, \quad C \Leftrightarrow q \supset q \wedge \neg p$$

(a) $A \Leftrightarrow B,$

(b) $\{A, B, C\}$ is satisfiable,

0 $A, B \models C,$

2 $(A \vee B)$ is valid.